Roll Number: _____	
Thapar Institute of Engineering and Technology Patiala	
Computer Science and Engineering Department **Auxiliary Examination**	
BE COE & CSE (Third Year)	**UCS701: Theory of Computation**
22nd February 2024, 5:30PM **Time: 3 Hours, Max Marks:100**	**Coordinator: Dr. Sunita Garhwal**

Note: Attempt all questions. Assume missing data, if, any, suitably.

Q1.	Consider the context-free grammar: S → ASB A → aAS | a | ε B → SbS | A | bb Convert the Context-free grammar into Chomsky Normal Form.	10
Q2.	Consider the following grammar and convert it into equivalent GNF grammar. S → XA | BB B → b | SB X → b A → a	15
Q3.	Prove that Regular languages are closed under union, complementation and intersection. Justify your answer with suitable examples.	15
Q4.	Given the context-free grammar: S -> AB | BC A -> BA | a B -> CC | b C -> AB | a Apply CYK algorithm to determine whether the string *w=baaba* belongs to the language generated by above given grammar.	15
Q5.	Consider the regular expression $r = b(a\|b)b^*b$ Convert the given r into NFA using Thompson construction. Convert the obtained NFA to DFA using Subset construction and minimized the obtained DFA.	10
Q6.	Prove that Context- free language are not closed under complementation.	10
Q7.	Draw the flowchart for the language L={$a^n b^n c^m$ | n, m ≥1}. Write down the transition diagram(PDA) for the above designed flowchart.	10
Q8.	Write down the logic for design of a Turing machine for the given language L={ ww | w∈{0,1} }. Design the Turing Machine for given L. Give the trace/ Instantaneous description for the string w=110110.	15